

Multiple Choice

1. **Answer:** A

Options: A) 10000 times more; B) 100 times more; C) 1000 times more; D) 10 times more

Note: Correct option: A - 10000 times more

2. **Answer:** C

Options: A) The velocity is the same on every point on the disk; B) The part closer to the axis has a longer period; C) The period of rotation is the same on every point on the disk; D) The part closer to the axis has a shorter period

Note: Correct option: C - The period of rotation is the same on every point on the disk

3. **Answer:** B

Options: A) Ability to magnify distant objects; B) Ability to measure the angular separation of objects; C) Ability to measure the distance between objects; D) Light-collecting efficiency

Note: Correct option: B - Ability to measure the angular separation of objects

4. **Answer:** C

Options: A) $m \cdot v$; B) $0.5 \cdot m / (R^3)$; C) $0.5 \cdot m \cdot (2 \text{ GM}/R)$; D) $0.6 \cdot G(M^2)/R$

Note: Correct option: C - $0.5 \cdot m \cdot (2 \text{ GM}/R)$

5. **Answer:** A

Options: A) 6000 K; B) 7000 K; C) 9000 K; D) 13000 K

Note: Correct option: A - 6000 K

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The path of the Sun in the sky throughout a year.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** A B and C. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key